

Uber Demand-Supply Gap Analysis

Data Analytics Internship Project Report

Prepared By

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Tools & Technologies Used

- Python (Pandas, NumPy, Matplotlib)
- PostgreSQL
- Microsoft Excel

Project Components

- Data Cleaning
- Exploratory Data Analysis (EDA)
- SQL Query Analysis
- Interactive Dashboard
- Business Insights & Recommendations

Submission Date

June 2026

Project Overview

Uber is one of the world's leading ride-hailing platforms that connects passengers with drivers through a mobile application. Efficient ride allocation is crucial for ensuring customer satisfaction and maintaining operational efficiency.

This project analyses Uber ride request data to identify demand-supply gaps, ride completion trends, cancellation patterns, and driver availability issues. The analysis was performed using SQL, Python, and Excel to generate meaningful business insights and recommendations.

Problem Statement

Many ride requests on the Uber platform are not successfully completed due to ride cancellations or the unavailability of drivers. These failed requests create a demand-supply gap, resulting in customer dissatisfaction and operational inefficiencies.

The objective of this project is to analyse ride request data and identify:

- Ride demand patterns
- Driver availability issues
- Cancellation trends
- Peak demand periods
- Demand-supply gaps across pickup locations and time slots

The insights generated from this analysis can help improve driver allocation and increase ride completion rates.

Dataset Information

The dataset contains ride request records collected from Uber's ride-hailing operations.

Dataset Summary

- Total Records: 6,745
- Total Columns: 11

Features Used

1. Request ID
2. Pickup Point
3. Driver ID

4. Status
5. Request Timestamp
6. Drop Timestamp
7. Request Hour
8. Weekday
9. Time Slot
10. Driver Assigned
11. Trip Completed

Data Quality Checks

The dataset was examined for missing values, duplicate records, and data consistency.

Key findings include:

- No duplicate records were found.
- Missing values in Driver ID and Drop Timestamp were business-related and expected.
- The dataset was considered suitable for further analysis.

SQL Analysis Findings

SQL queries were used to analyse ride request patterns, trip completion rates, cancellation trends, and demand-supply gaps within the dataset.

Key Metrics

Metric	Value
Total Ride Requests	6,745
Trip Completed	2,831
Cancelled Requests	1,264
No Cars Available	2,650
Completion Rate	41.97%
Failed Requests	58.03%

Query

Query History

156

-- Status distribution by time slot

157

select time_slot,status,count(*) as total_requests

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from uber_requests

159

group by time_slot, status

160

order by time_slot;

Data Output

Messages

Notifications

Showing rows: 1 to 18

Page No: 1

of 1

	time_slot character varying (30)	status character varying (50)	total_requests bigint
1	Afternoon	Trip Completed	491
2	Afternoon	No Cars Available	228
3	Afternoon	Cancelled	91
4	Early Morning	Trip Completed	681
5	Early Morning	Cancelled	668
6	Early Morning	No Cars Available	323
7	Evening	No Cars Available	1127
8	Evening	Cancelled	124
9	Evening	Trip Completed	642
10	Late Night	No Cars Available	299
11	Late Night	Cancelled	65
12	Late Night	Trip Completed	214
13	Morning	Trip Completed	404
14	Morning	No Cars Available	189
15	Morning	Cancelled	252
16	Night	Trip Completed	399

Figure 1: Status Distribution by Time Slot

Query

Query History

-- Total requests by pickup point

select pickup_point,
count(*) as total_requests
from uber_requests
group by pickup_point;

Data Output

Messages

Notifications



SQL

	pickup_point character varying (50)	total_requests bigint
1	City	3507
2	Airport	3238

Figure 2: Total Requests by Pickup Point

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-- PART 9: EXECUTIVE SUMMARY

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select pickup_point,

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count(*) as total_requests,

339

count(*) filter(where status = 'Trip Completed') as completed_trips,

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count(*) filter(where status = 'Cancelled') as cancelled_trips,

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count(*) filter(where status = 'No Cars Available') as no_car_requests,

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round(

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count(*) filter(where status = 'Trip Completed')::numeric /

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count(*) * 100,

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2

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) as completion_rate

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from uber_requests

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group by pickup_point;

Data Output

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Figure 3: Executive Summary Analysis

SQL Insights

- Only 41.97% of ride requests were successfully completed.
- More than half of the ride requests failed due to cancellations or driver unavailability.
- Driver unavailability was the primary reason for failed requests.
- Airport locations experienced significantly higher driver shortages.
- Evening and Early Morning periods recorded the highest ride demand.

The SQL analysis helped identify the primary factors contributing to the demand-supply gap and provided a foundation for further visualization and dashboard development.

Dashboard Analysis & Insights

An interactive Excel dashboard was created to visualize ride request trends, trip outcomes, demand-supply gaps, and driver availability patterns. The dashboard provides a consolidated view of key performance indicators and enables quick identification of operational bottlenecks.

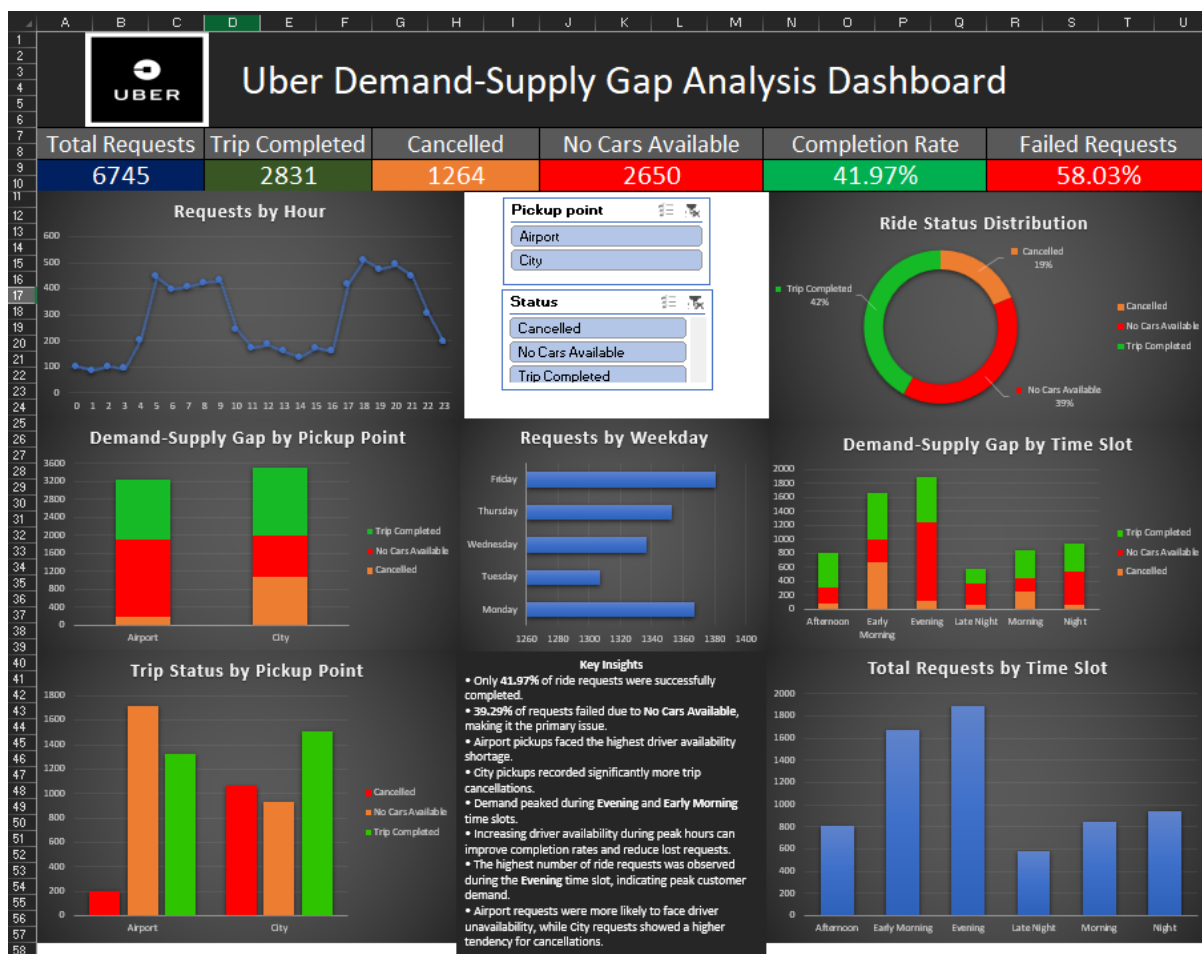


Figure 4: Interactive Excel Dashboard for Uber Demand-Supply Gap Analysis

Dashboard Insights

- A total of 6,745 ride requests were recorded in the dataset.
- Only 2,831 trips were successfully completed, resulting in a completion rate of 41.97%.
- Driver unavailability accounted for 2,650 failed requests, making it the largest contributor to ride failures.
- Airport pickup points experienced a significantly higher number of "No Cars Available" cases.
- City pickup points generated slightly more ride requests but also recorded a higher number of cancellations.
- Evening and Early Morning time slots showed the highest demand for rides.
- Peak demand periods experienced the greatest mismatch between ride requests and driver availability.

Business Interpretation

The dashboard highlights a clear demand-supply imbalance during peak hours. Improving driver availability during high-demand periods can reduce ride failures, improve customer satisfaction, and increase overall trip completion rates.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python libraries such as Pandas, NumPy, and Matplotlib to identify ride demand patterns, trip outcomes, driver availability issues, and demand-supply gaps.

The visualizations below highlight the key trends observed in the dataset.

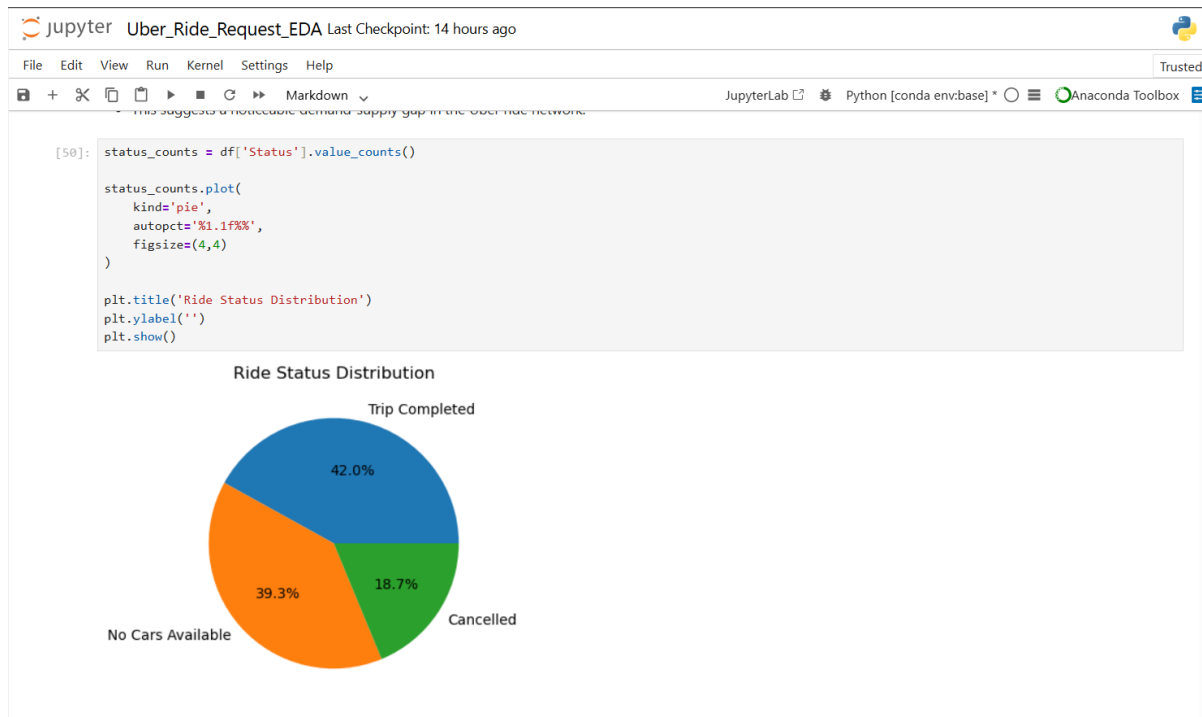


Figure 5: Ride Status Distribution

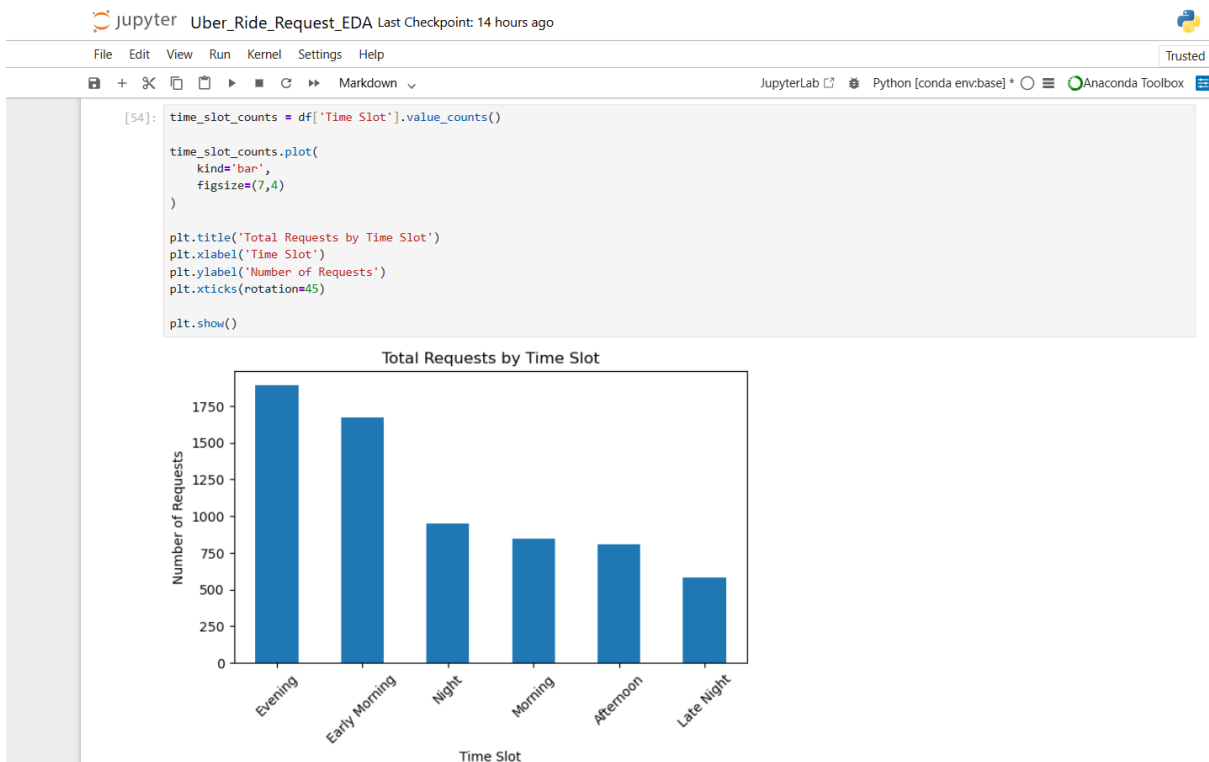


Figure 6: Total Requests by Time Slot



Figure 7: Trip Status by Pickup Point

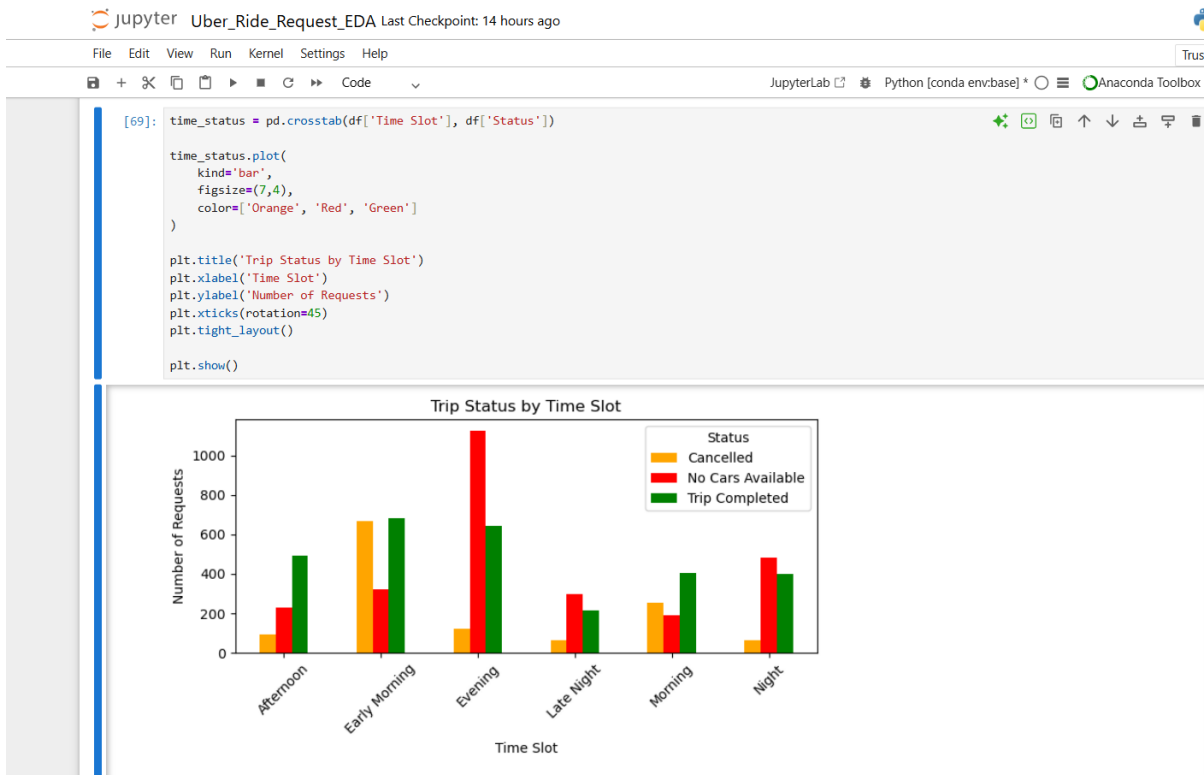


Figure 8: Trip Status by Time Slot

EDA Insights

- Only 42% of ride requests were successfully completed, while the remaining requests failed due to cancellations or driver unavailability.
- Evening and Early Morning periods recorded the highest ride demand across the dataset.
- Airport pickup locations experienced a significantly higher number of "No Cars Available" cases, indicating driver shortages.
- City pickup locations recorded more cancellations compared to Airport locations.
- Driver shortages were most prominent during Evening and Night time slots.
- Demand patterns remained relatively consistent throughout the weekdays.
- Peak demand periods showed the largest mismatch between customer requests and available drivers.

The EDA results support the findings obtained through SQL analysis and dashboard visualization, confirming that driver availability is the primary factor contributing to ride failures.

Key Findings & Recommendations

Key Findings

Based on the SQL analysis, Exploratory Data Analysis (EDA), and dashboard visualization, the following key findings were identified:

- The dataset contains 6,745 ride requests, out of which only 2,831 trips were successfully completed.
- The overall trip completion rate was 41.97%, indicating a significant demand-supply gap.
- Driver unavailability was the major reason for failed requests, accounting for 2,650 requests.
- Airport pickup locations experienced substantially higher driver shortages compared to City pickup locations.
- City pickup locations recorded a higher number of ride cancellations.
- Evening and Early Morning time slots generated the highest ride demand.
- Peak demand periods showed the largest mismatch between customer requests and available drivers.
- Ride demand remained relatively consistent across weekdays, suggesting that time of day had a greater impact than the day itself.

Recommendations

To reduce the demand-supply gap and improve ride completion rates, the following recommendations are proposed:

1. Increase driver availability during Evening and Early Morning peak demand periods.
2. Implement dynamic driver allocation strategies at Airport pickup locations.
3. Introduce incentive programs to encourage drivers to operate during high-demand hours.
4. Improve ride matching efficiency to reduce customer wait times and cancellations.
5. Use predictive analytics to forecast demand and proactively deploy drivers to high-demand areas.

Implementing these recommendations can help improve customer satisfaction, reduce ride failures, and enhance overall operational efficiency.

Conclusion

This project successfully analysed Uber ride request data using Python, PostgreSQL, and Microsoft Excel to identify demand-supply gaps and operational challenges.

The analysis revealed that driver unavailability was the primary factor contributing to ride failures, particularly during peak demand periods such as Evening and Early Morning time slots. Airport pickup locations were also found to experience significant driver shortages.

By leveraging data-driven insights, organizations can optimize driver allocation, improve ride completion rates, and enhance customer satisfaction. The findings and recommendations presented in this report can support better operational planning and decision-making for ride-hailing services.